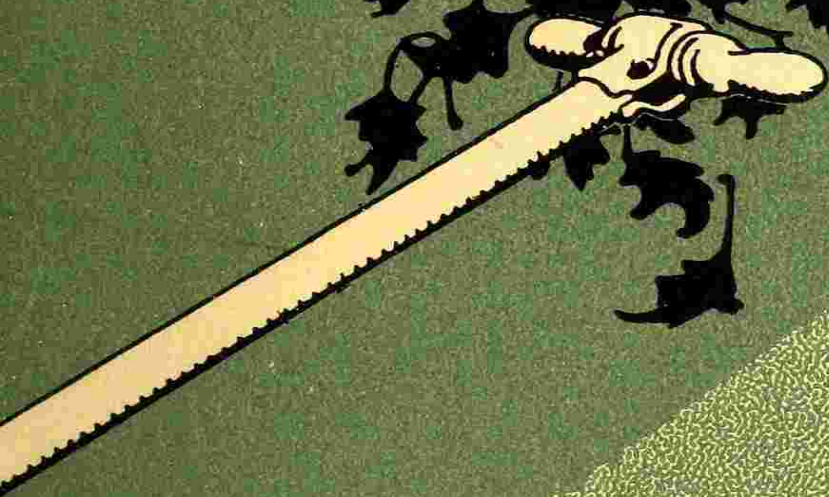
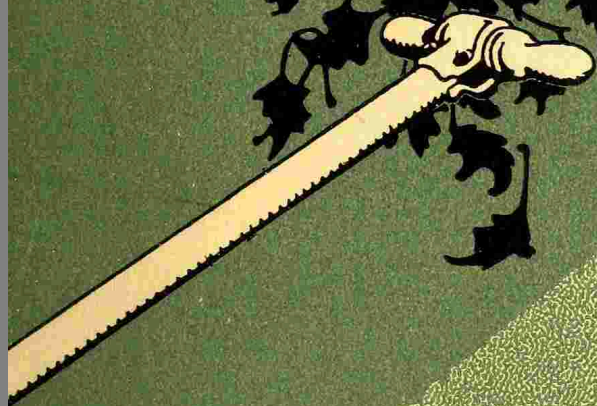


# KEEPING THE LAWN





# KEEPING THE LAWN



## **The Project Gutenberg eBook of Keeping the lawn**

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Keeping the lawn

Author: Leonard Barron

Release date: April 23, 2026 [eBook #78534]

Language: English

Original publication: Newburgh: Coldwell lawn mower co., 1921

Other information and formats: [www.gutenberg.org/ebooks/78534](http://www.gutenberg.org/ebooks/78534)

Credits: Bob Taylor, Charlene Taylor and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

\*\*\* START OF THE PROJECT GUTENBERG EBOOK KEEPING THE  
LAWN \*\*\*

# **KEEPING THE LAWN**

BY  
LEONARD BARRON

Editor The Garden Magazine

PUBLISHED BY  
COLDWELL LAWN MOWER CO.  
NEWBURGH, N. Y.

Copyright, 1921, by  
Coldwell Lawn Mower Company  
Newburgh, N. Y.



## KEEPING THE LAWN



### KEEPING THE LAWN



HE lawn is the dominant, characteristic feature of the American home garden. You may have the spread of green grass and nothing else and get a fairly pleasing setting for the house, but never a satisfying home from the latter without the lawn. It is almost a platitude, yet it is something to be considered by every home builder.

The lawn is indeed the essential base of the country, suburban or rural home. It distinguishes the “country” from the “city” with its paved yards and asphalted approaches. The lawn pleases the eye and it gives to the mind that sense of relaxation and repose which is looked for in all places away from the city itself. It has been well likened to the canvas on which the artist paints his picture; for when the landscape gardener comes to the problem of planting and embellishing the home grounds with an appropriate selection of trees, shrubs, and flowers his first consideration is the lawn, its place, extent, and the support it gives to the property as a whole. Without the lawn no garden worthy the name!

In view of the supreme importance of the lawn, then, an understanding of its upkeep is most desirable. And the delight of it is that there is nothing mysterious, nothing very onerous in keeping a lawn good.

Maintenance consists of a few simple details: rolling, cutting, watering, and feeding, and with a lawn properly made the most important of these is the cutting. In the old days this was done by a scythe, an effective instrument in the hands of the expert worker, but otherwise extremely unsatisfactory.

The invention of the lawn mower, less than a hundred years ago, marks one of those eras of progress that works revolution. Grass must be cut continuously throughout its growing season if the lawn is to retain the perfection of velvety evenness that is today within the possibility of every lawn owner, and this cutting can be accomplished easily in a definite way by means of the modern lawn mower, the credit for which in its present day form must be given to Thomas Coldwell, who somewhere about 1860 began his improvements on the original foreign-made model imported by Mr. H. W. Sargent for his extensive gardens at Fishkill, N. Y., directly across the Hudson River from Newburgh.

Thus was laid the foundation of an industry that has progressed by rapid strides to this day when there are available lawn mowers in a multitude of types and sizes, hand-operated, horse-drawn and motor driven, a size and style to fit every size and style of garden and lawn to the most extensive country estate, park, and golf course.

So accustomed are we of this day to see the well-mown lawn in the typical residential districts of our suburb and countryside that but little thought is given to the part played in the development of the lawn by the lawn mower having superseded the scythe.

Perhaps because the modern lawn mower does its work so well, —is such an efficient tool in operation, —there is a certain tendency among some gardeners to cut too close to the roots of the grass. Frequent mowings to keep the growth under proper control adds to the beauty of any lawn in keeping up a uniformly “well-kept” look. Irregular mowing leads to an unkempt irregular appearance of the turf because of the likelihood of putting off the job of cutting, on one pretext or another, until the grass has become unduly long.

When cut in that condition the exposed surface is brownish in color and thin looking—having none of the appearance of fresh grass.

The first cutting in spring as the season opens may well be deferred as long as possible in order to get a good growth started; and because in the early spring the uncertain fluctuations of the weather—from hot to cold and back again seemingly without any system—is likely to leave the newly exposed roots and lower parts of the grass stalks open to the burning rays of a sudden burst of extra strong heat.

But once the cutting is begun the work is to be kept up at frequent, fairly regular intervals to keep an uniform appearance to the grass. Cutting the grass has direct benefits, too, in that it prevents the grass plants from going to seed, and induces them to push up blades of fresh green in a dense mat that gives the so called desired velvety appearance.

Particularly is this desirable in the case of grass areas close around the house, on tennis or croquet lawns, and on grass edgings to walks, flower beds and borders.

After the first cutting in the season the grass will, as a rule, need cutting about once a week, and in some cases when the growing conditions are very favorable in the height of the season perhaps more frequently. As a matter of fact, judgment must be exercised here, the object being to keep the grass as nearly as possible at an uniform height.

The height of the cut is easily determined by a simple adjustment of the mower. About two inches is most satisfactory for the best interests of the grass itself—but a more well kept appearance may be desired and a much closer cut adopted. But the closer the cut the more feeding and other attention the lawn will need. But whatever height is determined, let the machine be set to that height and kept there, and regularly run over the lawn about once a week, and a constantly neat greensward is assured.

At the same time there need be no arbitrary “mowing day,” and a longer or shorter interval may be allowed according to circumstances. If for any reason the grass has been permitted to get unduly long it may be better to use a scythe and not cut very close, thus avoiding exposure of the under-texture of the turf.

The ideal day for cutting the grass is dull and cloudy—then the grass actually cuts more easily, and the little plants do not suffer any shock from the “pruning.”

With the knives all properly set the mower really keeps itself sharp while in use. The adjustment may be tested by inserting a piece of writing paper against the bottom knife and revolving the blades by hand, in this way testing the entire cutting surface. Improperly adjusted and blunt knives result in an irregular surface of the lawn or results in so pulling the little



grass plants as to loosen the roots. This may later on in a hot spell cause the death of the grass plant.

When using the lawn mower begin at one side of the lawn, pushing steadily forward at a brisk walking pace and traveling in a straight line to the opposite end of the lawn, thus making a clean straight strip of cut grass. Then return on a parallel strip with one of the guide wheels a couple of inches inside the previously cut strip. And so continue to the finish. This assures a perfect job and obviates leaving little uncut tufts here and there over the lawn which are certain to result from any irregular criss-cross manner of working. The next time of cutting start at the opposite end or at right angles.

Some lawn mowers have a provision for catching the clippings as they are thrown up by the revolving knives, when this is desired, and some way of gathering these clippings becomes necessary if the grass be allowed to become unduly long before it is cut. If there is no gathering box such clippings must be removed by raking, for which purpose a wooden rake is preferable to an ordinary garden rake. But as a general rule it will not be necessary or desirable to gather the clippings, but rather let them lie where they fall to act as a light mulch and protection to the grass. The short clippings which are likely to be the rule from a regularly cut lawn are thus distinctly beneficial to the grass plants. Only in extreme cases where the cuttings are very heavy should they be removed.

Next to cutting, rolling is the most important detail in keeping up the appearance of the lawn but, unlike cutting, rolling can generally be dispensed with after the season is well open. The purpose of rolling is to ensure a good contact between the roots of the grass plants and the soil in which they are growing, and there is a certain benefit from the use of a mower which is also equipped with a roller.

The degree of rolling of necessity will vary according to the soil, being greater in the case of a heavy soil than in one of a light, sandy nature. The action of alternations of frost and thaw during the winter results in loosening the roots of the plants in their hold on the soil and the effect of rolling is to compact this loosened mass of roots and soil and to maintain this condition so that as the plant grows it can draw its proper supply of nourishment from the ground.

In public parks and wherever great stretches of lawn exist the cutting is usually done by means of a machine of considerable size, horse drawn or motor driven. Such machines have rollers as integral parts and consequently the separate rolling operation may usually be dispensed with. But on the small hand-cut lawn preliminary rolling at the beginning of the season is a very important factor in its future welfare.

The only limit of weight of a roller that shall be used is the convenience of the person working it; in other words, the heavier the better. In actual service a three hundred pound roller is about the limit of a strong man's capacity, but a thousand pound weight would not be too heavy for the welfare of the lawn. The further advantage of the heavy roller is that apparently it prevents the establishment of the Crab Grass. A new lawn will need more rolling in its early stages than an old lawn.

Top dressing before rolling is useful in two ways; first, in helping to fill irregularities of the surface so as to restore a perfectly level stretch, and secondly, in actually carrying food supplies to the growing plants.

This top dressing may consist of finely sifted garden soil into which some bone dust has been mixed in the proportion of about four to one. This mixture may be lightly scattered over the surface by taking a shovelful at a time and distributing it by a broad sweep and then rolling it in where it falls.

Proper watering of the lawn is one of the least understood things in its upkeep if we would judge of general results seen in our suburban districts. The truth is that too many people approach the question of watering from the point of view of the picture a sprinkler makes rather than an understanding of the needs of the plants and their growing demands.

There is one simple guiding principle to be borne in mind whenever watering a lawn is considered. When water is given let it be given in sufficient quantity to literally give a thorough soaking. Too much emphasis cannot be put on this fact. Mere sprinkling which dampens the surface is not watering, and no matter how charmingly delightful the delicate spray of water glistening in the sunshine may be to the onlooker, it is in fact training the plants into a bad habit.

Of course, watering can be done by the sprinkling plan if one has assurance that sufficient water is allowed to flow—if the sprinkling

arrangement is allowed to stand sufficiently long in one place to absolutely soak the surface over which it plays.

Light sprinklings have a tendency to draw the roots of the plants upward in their search for moisture and when such watering is done early in the season and the season itself is dry, the plants so treated fall easy victims to a spell of hot sunshine during midsummer drought.

There can be no calendar of watering, no time-table on which to work. Watering the lawn, as in all other garden operations, is a matter of adjustment to conditions. It will vary with the soil, with the sub-soil, and with the situation. It may be necessary to begin watering in April some years, but in others it will hardly be necessary to water at all.

All surface sprinklings do, however, accomplish something; in that they check transpiration of water from the grass leaves and from the surface of the ground for a short time, and in that way sprinkling in very hot weather may be palliated but must be followed up as soon as possible by an application of water that will be thorough in every sense of the word.

The lawn made on a well prepared soil will not need as much water as one made on a poor soil or one that is of insufficient depth. A lawn perfectly prepared, made on good soil, of good depth, and not unduly drained should hardly need water at all; but the fact is that most of us have to take our lawns as we find them and on light shallow soils watering is a real necessity.

When water is supplied by the hose remove the nozzle, lay the open end of the hose on the ground, and allow the water to flow freely in one spot about an hour. Be careful that there is not too much pressure. If there be any danger of the flow of water wearing a channel into the soil place a piece of board on the lawn, laying the open end of the hose on this to distribute the water. When the lawn does need water give it day or night or both.

There is no real foundation for the fear sometimes expressed of watering in the sunshine. The only possible injury that could result is a slight burning of some of the tender tips through the sun's rays being focused through pendant drops of water as in a burning glass. But such injury would not be any more noticeable than the browning and shrivelling of the cut edge of a blade of grass that occurs whenever the lawn mower is passed over it and

that is so small as to be hardly recognizable in ordinary circumstances. Water whenever and wherever it is needed. Better the welfare of the entire plant than the temporary welfare of an individual leaf!

As the lawn ages its food supply must be maintained. Obviously, the right way to keep the lawn green is to so make it in the first place that it has all its necessary stock of food in the soil for a number of years; but surface dressings may be used to maintain growth. Corn cannot be expected from an unfed land and yet corn land is cultivated annually. Grass is a permanent crop enduring for many years and really needs richer feeding than the vegetable garden. But does it get it? It does not, because of the tendency of weak humanity to take a chance on the present and defer trouble as long as possible.

After the lawn is once made it can be fed only through surface dressings and the lighter the soil and more sandy, the greater the demand for fertilizers. On such soils dressings of well decayed stable manure will give better results than anything else. On heavy soils chemical fertilizers are equally satisfactory. The latter have the advantage, and great advantage, of being free from weeds. Stable manure nearly always carries in weed seeds. This top dressing may be put on in the fall as a mulch or it may be spread on any time during winter, even when the ground is covered with snow.

Tobacco stems are often used, and being free from weed seeds, answer the purpose very well. Pulverized sheep manure as sold from the stock yards is also free from weed seeds and may be used with impunity.

Chemical fertilizers are cleaner in appearance and, in fact, have more concentrated food value. Nitrate of soda has long been a standby scattered over the surface of the lawn at the rate of one to two thousand pounds to the acre. (A plot a hundred by twenty-five feet is about a fifteenth of an acre). This may be applied just in advance of a rain or it may be first of all dissolved in water and applied in solution, which is perhaps the more convenient way, using one pound of soda nitrate to forty gallons of water.

Ammonium sulphate, another source of nitrogen, may also be used, and it is the experience of most observers that where chemicals are used better results are obtained by alternating these two chemicals.



In order to keep up the general appearance of the lawn occasional top dressings to fill in depressions or holes may be made any time of the year by good sifted soil from the garden in which some bone meal has been mixed. Lawn seed is then scattered on the surface and the whole rolled or beaten down.

In places where wood ashes are available an excellent lawn fertilizer is at hand and may be used in the same manner as the chemical fertilizers.

Lawns made from the recognized lawn mixtures contain a great quantity of blue-grass, which will not grow on a soil showing an acid reaction. Therefore, wood ashes are valuable because they supply an alkaline element.

Lime, that is, agricultural or slaked lime, two bushels to a plot one hundred by twenty-five, may be used on any lawn and will help to maintain an alkaline soil condition which is essential to the production of that blue-green color of the lawn that is the acme of perfection of lawn color.

Unfortunately, weeds will creep into good lawns and eternal vigilance is needed to keep them in control. On an acid soil we may have a yellow, poor looking lawn without weeds; or a perfect, blue-green, rich looking stretch of grass with weeds, for, curiously, the alkaline condition that makes for good grass also encourages the weed plants.

To suppress them and keep them out requires constant vigilance. Broad-leaved weeds can be destroyed by scattering sulphate of iron, which being caught on the broad, outstretched leaf, practically burns it and in time kills the plant (it does not affect the grass because of the upright habit of growth); but the more satisfactory way is to keep the weeds down by constantly cutting out such things as dandelions and rib-grass, as quickly as they appear, then making good any hole in the surface and reseeding the bare spot. Dandelions can be kept under control by preventing flowering.

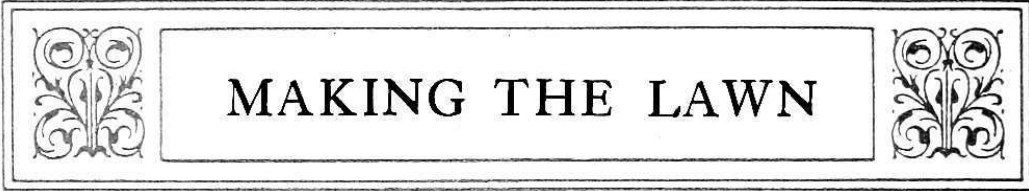
Crab-grass, the pest of very many lawns, is an annual, the seeds of which are carried in by means of birds, chiefly sparrows. Where it appears it must be bodily raked up, as the lawn mower will not cut it, since it creeps along on the surface of the ground below the cut level of the well kept lawn. On account of its manner of growth it must be literally grubbed up by a sharp iron rake. Badly infected lawns should be dug over and remade.

The presence of moss on the lawn indicates insufficient under drainage and is to be remedied only by putting in proper drainage. Surface dressings of lime may give temporary relief, but nothing more.

While it is better to dig out rank growing weeds in the fall rather than in the spring, still it is better to do it in the spring than not at all. Constantly reseeding the surface of the lawn with a high grade lawn seed mixture will do much towards keeping out the weeds because of the mathematical fact that two things cannot occupy the same place at the same time. The best way to keep out the weeds is to keep in the grass.

Chick weed is best raked out before it is permitted to flower. It can also be controlled by dusting on sulphate of iron.

Brown patches on the lawn may be produced by different causes; one is a fungus which is on the surface and may be controlled by dusting dry Bordeaux mixture on the affected areas. Another source of the brown patch is the working of an underground grub that cuts through the roots, loosening sheets, as it were, of dead turf which may be rolled up. This grub, when not very severe, can be controlled by gasoline which will kill it in the ground. In very severe cases the lawn must be remade after having been plowed and allowed to stand over one winter open to frost. This grub is a large white, fat, sluggish creature, curved into a crescent shape.



## MAKING THE LAWN

### MAKING THE LAWN



MAKING THE LAWN. The making of a good lawn depends on the making of a proper foundation. A deep soil such as will make a good vegetable garden is ideal for the lawn.

If the land is very light and sandy, stable manure or other vegetable matter must be introduced. The soil can hardly be too rich or too good for a good lawn.

In a wet situation drain tile, the porous agricultural kind, must be put in to take off surplus water and help aerate the land.

With a proper foundation the surface must be broken up, pulverized, leveled, smoothed and raked over until all large bits of rubbish, pebbles, stone, and other foreign matter have been removed. This will result in the making of a good seed bed in which the seed may be sown.

The surface need not be flat like the top of a table, a rolling contour, indeed, gives a more natural sweep.

Buy the best grass seed that the dealers offer and buy it by weight, twenty pounds to the bushel. In making a new lawn use five bushels to the acre. When subsequent applications of top dressing are made each year, one to two bushels to the acre may be used. Some people prefer to use three bushels to the acre in making a new lawn which, while it will produce a lawn, leaves the way open for weeds and does not give so fine a texture to the grass.

Kentucky blue-grass is the recognized lawn grass. It, alone, has that desirable blue-green color that gives a perfect lawn. It is slow to germinate and to establish itself, therefore the practical gardener uses a mixture

containing other grasses in order to establish a greensward quickly, while the blue-grass is developing; and where the proper soil conditions prevail—alkalinity—the blue-grass will eventually crowd out most other grasses.

Each seedsman has his own particular formula for his lawn mixture. In practice they differ in detail but not in principle, and generally contain about thirty per cent. Kentucky blue-grass, thirty per cent. Red-top, and the balance being made of meadow fescue, crested dog-tails, English rye, and perhaps a few other grasses, varying according to circumstances.

The rye grass germinates in a few days and quickly gives a green appearance to the new lawn.

Kentucky blue-grass must have an alkaline soil. Rhode Island bents and meadow fescue are suited to acid and seaside soils.

The actual use of mixtures is a much debated question and may vary in different places; and in practice the lawn maker will be well-advised to take the highest grade lawn mixture offered by reliable seedsmen. These mixtures are varied in their composition somewhat for shaded places or for clay soils, and rather than make these different mixtures themselves, the lawn maker would be better advised to place reliance upon an established seedsman and use the branded mixtures which are the result of long experience and observations.

For the information of those who may desire to make their own mixtures, or have them made up to order, the following proved formulas are given:

On soils of average fertility and composition, this has given satisfaction:

|                   |           |
|-------------------|-----------|
| Kentucky Blue     | 10 quarts |
| Rhode Island Bent | 8 quarts  |
| English Rye       | 3 quarts  |

This is a crude, although reliable mixture. Indeed, it may be called a lawn mixture reduced to its simplest elements. The quantities given in quarts are based on thoroughly recleaned seeds.

A more refined mixture, including a fancy Red-top for filling in during the early years of the lawn and after the English Rye has lived its life, is as follows:

|                           |         |
|---------------------------|---------|
| Fancy Kentucky Blue-grass | 10 lbs. |
| Fancy Red-top             | 4 lbs.  |
| R. I. Bent                | 3 lbs.  |
| English Rye               | 3 lbs.  |

This formula is expressed in weight and may be used as a fair basis of comparison with the preceding formula which is expressed in bulk.

This mixture would give twenty pounds to the bushel, and would be sufficient for one-fifth of an acre, say about eight thousand square feet. Fancy seed is specified in the formula both as regards the blue-grass and the red-top. The twenty pounds weight of this mixture, designated in the trade as a bushel, would, of course, not fill the actual measured bushel.





\*\*\* END OF THE PROJECT GUTENBERG EBOOK KEEPING THE  
LAWN \*\*\*

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

# THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at [www.gutenberg.org/license](http://www.gutenberg.org/license).

## **Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works**

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website ([www.gutenberg.org](http://www.gutenberg.org)), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

## 1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to



you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

## **Section 2. Information about the Mission of Project Gutenberg**

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at [www.gutenberg.org](http://www.gutenberg.org).

### **Section 3. Information about the Project Gutenberg Literary Archive Foundation**

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at [www.gutenberg.org/contact](http://www.gutenberg.org/contact)

### **Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation**

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

## **Section 5. General Information About Project Gutenberg electronic works**

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:  
[www.gutenberg.org](http://www.gutenberg.org).

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.